

Executive Cost & Sourcing Overview: This document provides the formal itemized Bill of Materials (BoM) for the Giri-Drishti 2-axis direct-drive counter-UAS gimbal platform. Hardened for high-altitude operation (5,000 m ASL / 540 mbar / -55°C), the baseline platform replaces active resistive heating with passive athermalization and active observer control, achieving an **88% reduction in total thermal power consumption**.

Financial Summary & Currency Conversion:

- **Total Prototype Cost:** ₹6,41,698 INR (\$7,685.00 USD)
- **Reference Exchange Rate:** 1 USD (\$) = ₹83.50 INR
- **Highest Cost Subsystem:** Optomechanical Yoke Chassis (SiC_p/Al MMC Composite) @ ₹1,54,475 INR (24.1% of total budget)

Item	Subsystem Component	Manufacturer / Part No.	Material / Spec	Qty	Unit Cost (₹)	Total Cost (₹)	High-Altitude Hardening Feature
1.0	Gimbal Yoke Chassis	Custom Machined	SiC _p /Al Composite (65% SiC)	1	₹1,54,475	₹1,54,475	Low CTE (6.5e-6/K), high stiffness at -55°C
2.1	Azimuth Drive Motor	TQ-RoboDrive ILM-70x18	Frameless PMSM Direct Drive	1	₹1,00,200	₹1,00,200	Vacuum/low-pressure rated winding insulation
2.2	Elevation Drive Motor	TQ-RoboDrive ILM-50x14	Frameless PMSM Direct Drive	1	₹79,325	₹79,325	Zero-backlash direct coupling
3.1	Azimuth Encoder	RLS AksIM-2	19-bit Absolute Magnetic Ring	1	₹56,780	₹56,780	Non-contact, immune to condensation/ice
3.2	Elevation Encoder	RLS AksIM-2	19-bit Absolute Magnetic Ring	1	₹56,780	₹56,780	High shock/vibration tolerance (MIL-STD-810H)
4.0	Flight Controller	Custom STM32F767 / TI C2000	4-Layer Polyurethane Coated	1	₹35,070	₹35,070	1.48x IEC 60664-1 PCB trace gap (0.67mm)
5.0	Power Drivers	GaN Systems GS66508B	650V GaN HEMT Bridges	2	₹25,885	₹51,770	3.19x dielectric safety factor @ 540 mbar
6.0	Breathing Vent Valve	Donaldson ePTFE Membrane	IP67 Hydrophobic Vent	2	₹3,758	₹7,515	Continuous pressure equalization at 5,000m ASL
7.0	Rotary Shaft Seals	Trelleborg Turcon Variseal	Fluorosilicone / PTFE Spring	2	₹7,098	₹14,195	Rated to -65°C, low stiction grease
8.0	Precision Bearings	Kaydon Reali-Slim	440C Stainless / MIL-PRF-23827	4	₹20,040	₹80,160	Low-temp grease prevents sub-zero lockup
9.0	Thermal Insulation	Aspen Aerogel Pyrogel XTE	3mm Flexible Aerogel Blanket	1	₹5,428	₹5,428	k = 0.015 W/m-K thermal protection

GIRI-DRISHTI C-UAS GIMBAL PLATFORM

CONFIDENTIAL / DEFENSE SPEC

Itemized Bill of Materials (BoM) — High-Altitude Environmental Hardening Specifications

TRL-4 PRE-PROTOTYPE BASELINE

Item	Subsystem Component	Manufacturer / Part No.	Material / Spec	Qty	Unit Cost (₹)	Total Cost (₹)	High-Altitude Hardening Feature
TOTAL	Complete Platform Prototype	Giri-Drishti C-UAS Gimbal	TRL-4 Pre-Prototype Assembly	-	-	₹6,41,698	88% power savings vs 320W active heater COTS